LAB PROGRAMS (21-25)

COMPUTER VISION-ITA-0505

SLOT-D

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QUESTION-21

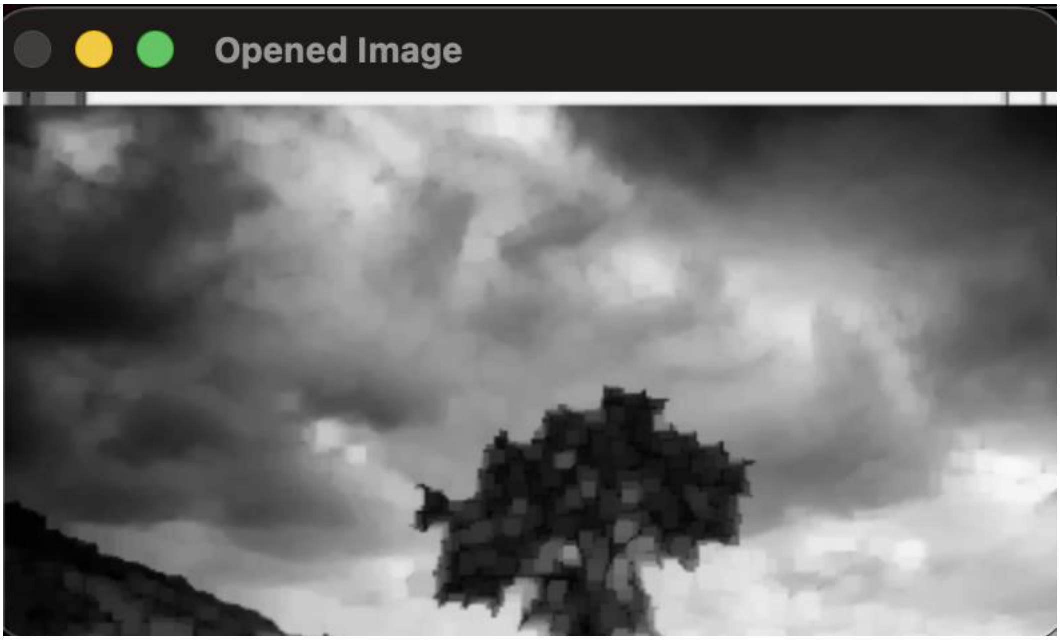
Implement the Opening technique as a Morphological operation to dilate the foreground regions based on Open CV.

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if the image is loaded
if image is None:
   print("Error: Image not found!")
   exit()
# Convert image to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Display original image
cv2.imshow("Original Image", image)
# Display grayscale image
cv2.imshow("Gray-scale Image", gray_image)
```

```
# Wait for a key press
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-22:

Implement the Closing technique as a Morphological operation to dilate the foreground regions based on Open CV.

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
```

```
 # Check if the image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Apply Gaussian Blur
blur_image = cv2.GaussianBlur(image, (15, 15), 0)
# Display original image
cv2.imshow("Original Image", image)
# Display blurred image
cv2.imshow("Gaussian Blur Image", blur_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:

QUESTION-23:

implement the Top hat technique as a Morphological operation to dilate the foreground regions based on Open CV

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if image is loaded
if image is None:
    print("Error: Image not found!")
    exit()
# Convert to grayscale
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply Canny edge detection
edges = cv2.Canny(gray_image, 100, 200)
# Display images
cv2.imshow("Original Image", image)
cv2.imshow("Edge Detected Image", edges)
# Wait for key press
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:

QUESTION-24:

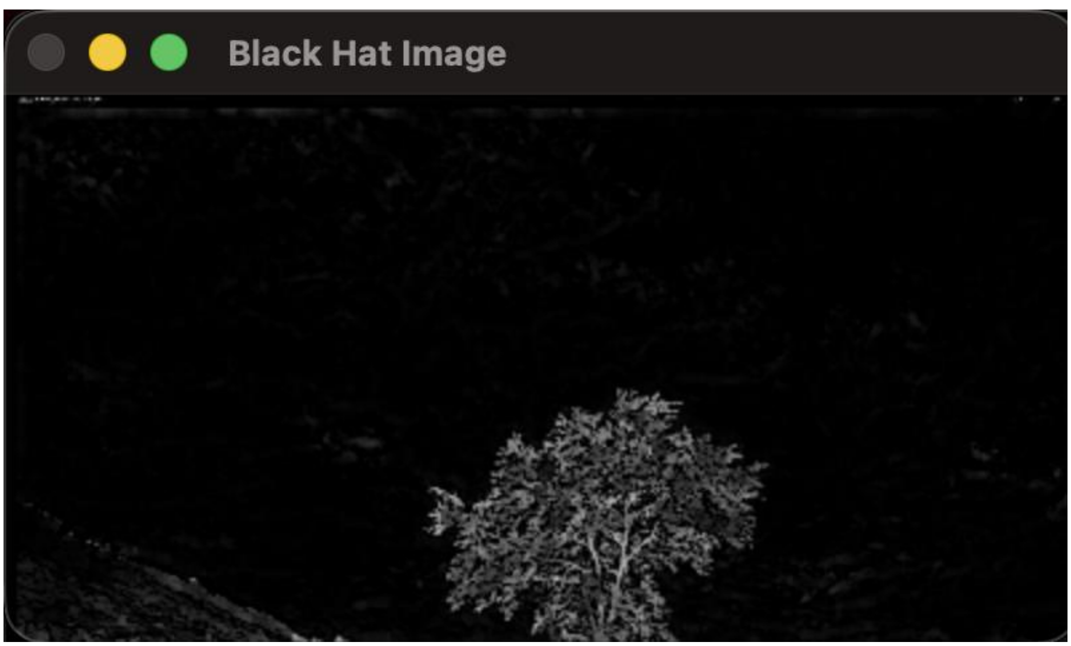
Implement the Black hat technique as a Morphological operation to dilate the foreground regions based on Open CV.

PYTHON CODE:

```
import cv2
# Read the image
image = cv2.imread("sample.jpg")
# Check if image is loaded
if image is None:
    print ("Error: Image not found!")
    exit ()
# Convert to grayscale
```

```
gray_image = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
# Apply Histogram Equalization
equalized_image = cv2.equalizeHist(gray_image)
# Display images
cv2.imshow("Original Grayscale Image", gray_image)
cv2.imshow("Histogram Equalized Image", equalized_image)
# Wait until a key is pressed
cv2.waitKey(0)
# Close all windows
cv2.destroyAllWindows()
```

OUTPUT:



QUESTION-25:

Recognize watch from the given image by general Object recognition using

Open CV.

PYTHON CODE:

```
import cv2
import matplotlib.pyplot as plt
def analyze_histogram(image_path):
    # Read the image
    image = cv2.imread(image_path)
    # Check if image is loaded
    if image is None:
        print("Error: Image not found!")
        return
    # Display the original image
    cv2.imshow("Original Image", image)
    # BGR color channels
    colors = ('b', 'g', 'r')
    # Create histogram plot
    plt.figure(figsize=(8, 5))
    for i, color in enumerate(colors):
        histogram = cv2.calcHist([image], [i], None, [256], [0, 256])
        plt.plot(histogram, color=color, label=f"{color.upper()} Channel")
    plt.title("Color Histogram Analysis")
    plt.xlabel("Pixel Intensity")
    plt.ylabel("Number of Pixels")
```

```
    plt.xlim([0, 256])
    plt.legend()
    # Show histogram
    plt.show()
    # Wait for key press
    cv2.waitKey(0)
    cv2.destroyAllWindows()
# Call the function
analyze_histogram("sample.jpg")
```

OUTPUT: